

RASPBERRY PI NAS – USB 3.x TO NVMe BRIDGE

Board target: 90 mm x 90 mm, 4 layers. SCHEMATIC ONLY – NO PCB LAYOUT.

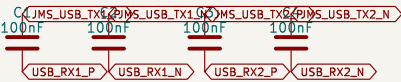
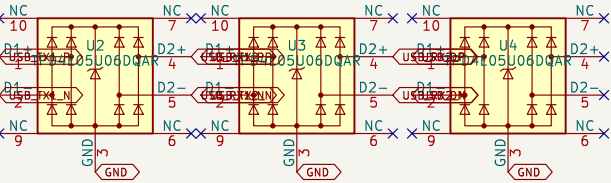
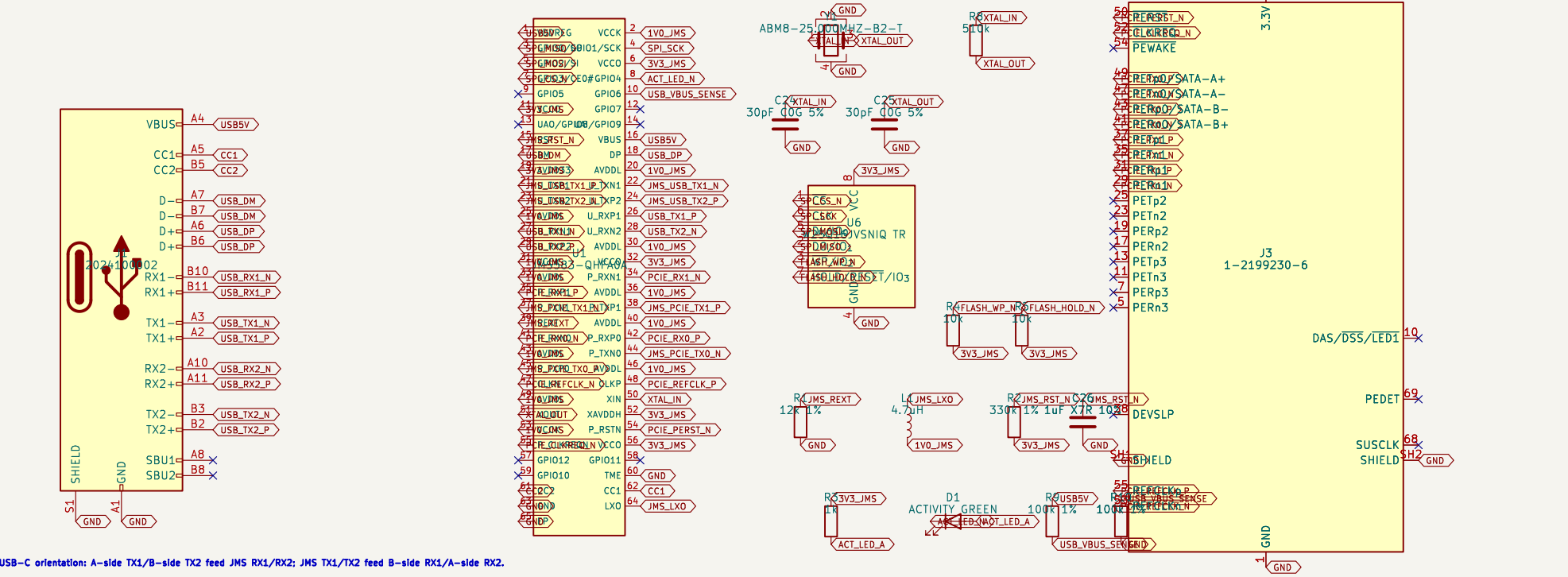
1. USB INTERFACE

2. JMS583 BRIDGE

3. NVME / M.2

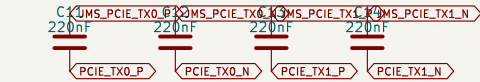
JMS583 CORRECTIVE REVIEW APPLIED – SEE schematic_review.md

1V0 core/analog rail; EP and TME grounded; 330k/1uF reset; GPIO4 active-low LED; calculated 30pF crystal loads.



M.2 M-Key – 2280 NVMe – PCIe x2 electrically

Lanes 2 and 3 intentionally unconnected. Connector pin names follow host-centric PET/PER convention.



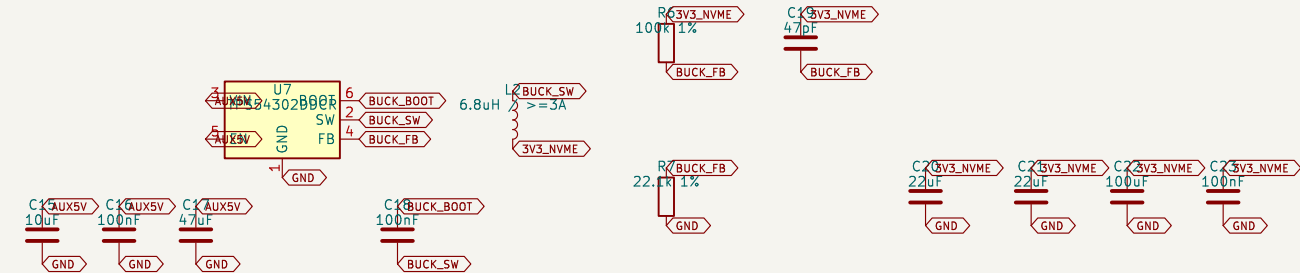
4. POWER

SPLIT POWER: USB5V -> JMS583; AUX5V -> NVMe 3V3 buck; COMMON GND

AUX5V and 3V3_NVME must be valid before USB enumeration when SSD access is required.



AUX5V INPUT – CONNECTOR TBD



TP554302: 6.8 uH, 44 uF output, 100k/22.1k divider and 47 pF feed-forward follow TI's 3.3 V table.

TP554302 is limited to 3 A maximum; verify SSD peak/transient demand before release.

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File: raspberry_pi_nas_nvme.kicad_sch

Title:

Size: A3

Date:

Rev:

KiCad E.D.A. 9.0.5

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